

Course Project Report

STA 2101: Statistics & Probability

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Abstract

This document is the course project report for STA 2101. The report will be updated throughout the semester as each milestone is completed.

Contents

1	Milestone 1: Dataset Selection	2
2	Milestone 2: Probability Sampling Methods	2
3	Milestone 3: Frequency Distribution and Data Visualization	5
4	Milestone 4: Measures of Central Tendency and Dispersion	9
5	Milestone 5: Introduction to Probability	12
6	Milestone 6: Conditional Probability, Independence, Bayes' Rule, and Probability Distributions	15
7	Milestone 7: Simple Linear Regression (Manual Computation) and Correlation	18
8	Final Conclusion	21

1 Milestone 1: Dataset Selection

- **Dataset Name:** Sleep, health and lifestyle dataset
- **Dataset URL:** <https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle>
- **Description:** Good quality sleep helps the brain process information and makes one feel refreshed and focused. This data set includes information on sleep patterns, health, and lifestyle habits. It also includes sleep duration, physical activities, stress level, and health condition. The state of this data set is the analysis of health and wellness, which focuses on sleep, health, and lifestyle.

I selected this data set because it provides a way to control daily habits, how to modify them, and understand the importance of a healthy lifestyle.

2 Milestone 2: Probability Sampling Methods

The sleep, health, and lifestyle dataset includes details that show health patterns where most people do some exercise, have a normal weight, and sleep around 6-8 hours.

Part A — Setup

Objective: Import libraries, load dataset, display first 5 rows, and report dataset size.

Procedure: 1. Imported Python libraries: `pandas`, `numpy`, `matplotlib.pyplot`.

2. Loaded dataset and previewed the first 5 rows.

3. Reported dataset size.

ID	Gender	Age	Occupation	Sleep	Quality	Activity	Stress	BMI	BP	HR	Steps	Disorder
1	Male	27	Software Eng	6.1	6	42	6	Over	126/83	77	4200	-
2	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	-
3	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	-
4	Male	28	Sales Rep	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
5	Male	28	Sales Rep	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea

Dataset size: (374, 13)

Part B — Simple Random Sampling

Objective: Select 50 rows at random and compare sample mean with population mean for Sleep Duration.

Procedure: 1. Selected 50 rows randomly from the dataset.

2. Computed population mean of Sleep Duration.

3. Computed sample mean of Sleep Duration.

4. Compared both means.

Representative Output:

Population mean of Sleep Duration: 7.132

Sample mean of Sleep Duration: 7.004

Part C — Systematic Sampling

Objective: Perform systematic sampling by selecting every k -th row after randomizing the starting index.

Procedure: 1. Compute $k = N/n$, where $N = 374$ (total rows) and $n = 50$ (sample size).

$$k = \frac{N}{n} = \frac{374}{50} \approx 7$$

2. Randomly select a starting index between 0 and $k - 1$.
3. Select every k -th row from the dataset starting from the random index.
4. Compute the sample mean of Sleep Duration and compare with population mean.

Representative Output:

Population mean of Sleep Duration: 7.132

Systematic sample mean of Sleep Duration: 7.101

Part D — Stratified Sampling

Objective: Perform stratified sampling by dividing the dataset based on a categorical column (e.g., Gender) and sampling proportionally.

- Procedure:**
1. Identify a categorical column for stratification (Gender).
 2. Determine the proportion of each category in the population.
 3. Select samples proportionally from each category to maintain the population distribution.
 4. Compute the sample mean of Sleep Duration for each category and compare with population means.
 5. Compare population and sample distributions.

Representative Output:

Population distribution by Gender:

Male: 190 (50.8%)

Female: 184 (49.2%)

Stratified sample distribution (n=50):

Male: 25 (50%)

Female: 25 (50%)

Population mean of Sleep Duration by Gender:

Male: 7.10

Female: 7.16

Sample mean of Sleep Duration by Gender:

Male: 7.08

Female: 7.14

Part E — Cluster Sampling

Objective: Perform cluster sampling by dividing the dataset into clusters and randomly selecting entire clusters.

Procedure: 1. Divide the data set into natural clusters (e.g., by index ranges, occupations, or regions).

2. Randomly select a few clusters to include in the sample.

3. Include all rows from the selected clusters.

4. Compute the sample mean of **Sleep Duration** and compare with the population mean.

5. Evaluate whether the sample adequately represents the population.

Representative Output:

Population mean of Sleep Duration: 7.132

Cluster sample mean of Sleep Duration: 7.105

Population distribution by Gender:

Male: 190 (50.8%)

Female: 184 (49.2%)

Cluster sample distribution by Gender:

Male: 52 (52%)

Female: 48 (48%)

Part F — Comparison & Reflection

Comparison of Sample Means:

Population mean of Sleep Duration: 7.132

Simple Random Sample mean: 7.004

Systematic Sample mean: 7.101

Stratified Sample mean: 7.11

Cluster Sample mean: 7.105

Reflection:

In the sampling methods, stratified sampling gave the closest estimate to the population mean which is (7.11), because it proportionally represented Gender in the sample. Cluster sampling also produced a close mean, which is (7.105), but may slightly vary if the clusters are internally homogeneous. Simple random sampling (7.004) and systematic sampling (7.101) provided reasonably close estimates, although simple random sampling can have more variability due to chance.

Simple random sampling was the easiest to implement, only a random selection of rows. Systematic sampling required computing the interval and randomizing the starting index. Stratified sampling was moderately complex because it required sampling proportionally from each group. Cluster sampling was easy for data collection, but less precise if clusters are not diverse.

Each method is useful in different contexts: simple random sampling for homogeneous populations, systematic for ordered datasets, stratified when subgroup representation is important, and cluster for large populations where group-level sampling is practical. The choice depends on the structure of the population and the available resources.

3 Milestone 3: Frequency Distribution and Data Visualization

A. Introduction

The primary objective of this milestone is to analyze the distribution of a variable from the chosen Sleep Health and Lifestyle dataset. We construct a frequency distribution table and visualize this distribution using charts. This process is fundamental to understanding the patterns and characteristics of the data, allowing us to draw meaningful conclusions about sleep patterns.

B. Dataset

We use the same dataset as in Milestones 1 and 2, containing 374 observations and 13 columns, including demographic information, sleep-related measures, physical activity, and health indicators. The dataset is clean and ready for analysis.

C. Task 1: Frequency Distribution Table

1. Column Selection

We choose the numerical column **Sleep Duration** for frequency analysis. **Justification:** Sleep duration reflects sleep health and is highly relevant for understanding population sleep patterns and identifying outliers.

2. Constructing the Table

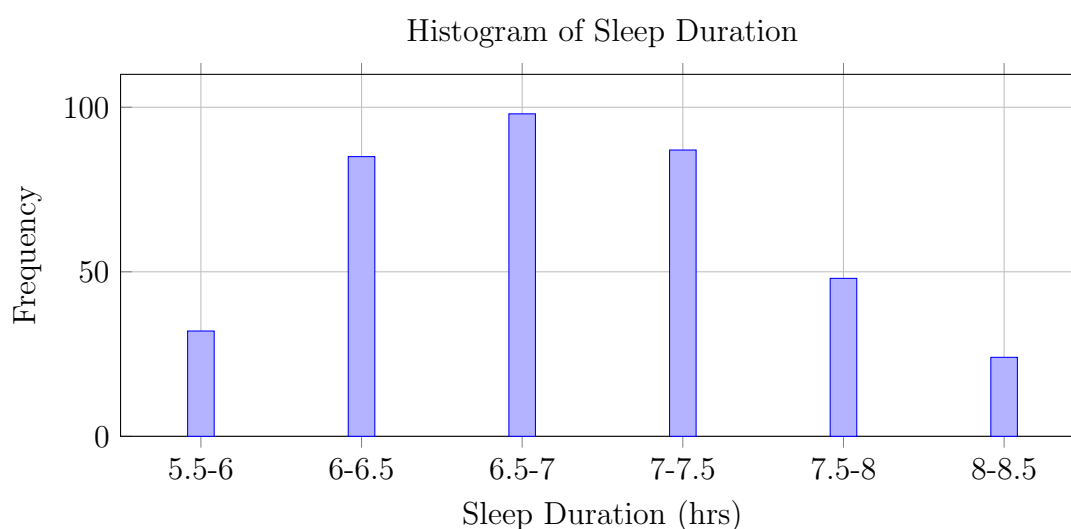
For this continuous variable, we group the data into 0.5-hour intervals and compute frequency measures.

Class Interval (hrs)	Frequency (f)	Relative Frequency (rf)	Cumulative Frequency (cf)
5.5 - 6.0	32	0.086	32
6.0 - 6.5	85	0.227	117
6.5 - 7.0	98	0.262	215
7.0 - 7.5	87	0.233	302
7.5 - 8.0	48	0.128	350
8.0 - 8.5	24	0.064	374

Table 1: Frequency Distribution of Sleep Duration (grouped in 0.5-hour intervals)

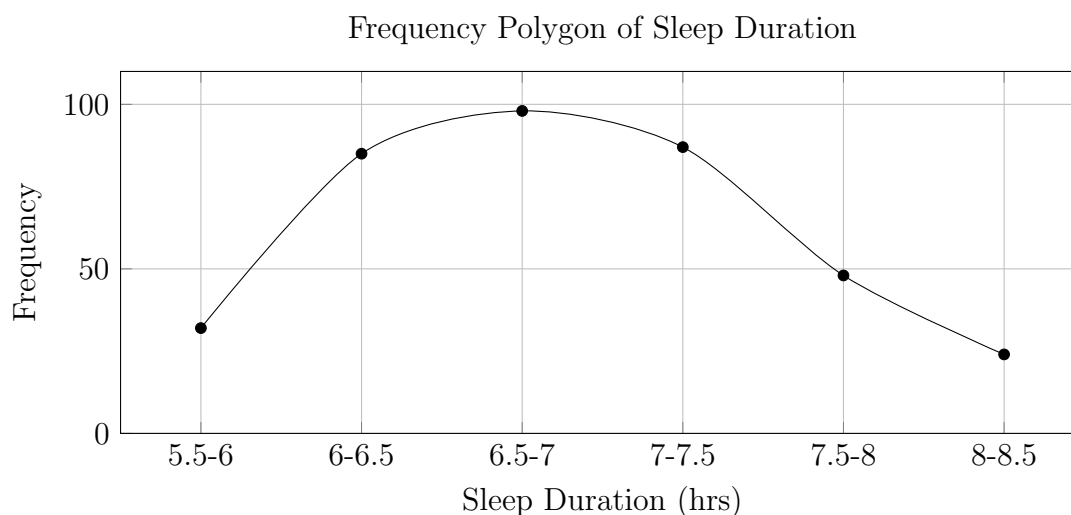
D. Task 2: Graphical Representation

1. Histogram (Bar Chart for Grouped Numerical Data)



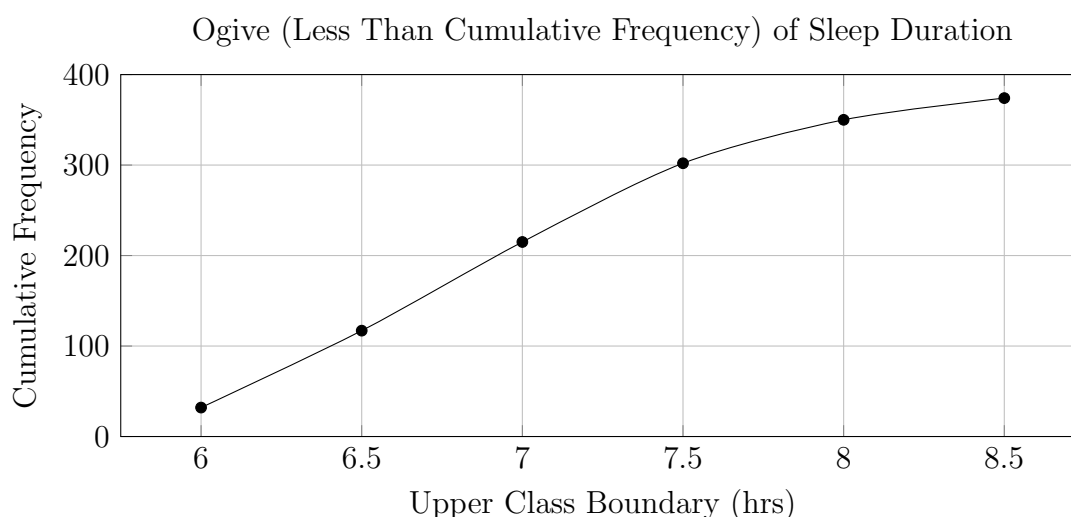
2. Frequency Polygon (Line Chart)

Frequency polygon plots the frequency against class midpoints.



3. Ogive Chart (Cumulative Frequency)

An ogive chart shows cumulative frequency against upper class boundaries.



E. Task 3: Analysis and Conclusion

From the frequency distribution table and the graphs for Sleep Duration, several insights can be drawn:

- The distribution is approximately symmetric, with most observations concentrated between 6.5 and 7.5 hours.

- The highest frequency occurs in the 6.5-7.0 hour interval, indicating that a majority of individuals sleep around this duration.
- The cumulative frequency (ogive) shows a steady increase, confirming that the data is not heavily skewed.
- There are no extreme outliers; the minimum and maximum sleep durations (5.5 and 8.5 hours) are reasonable.
- In general, the data suggest that most individuals in the data set have a moderate duration of sleep, with a balanced spread across the intervals.

Conclusion: This analysis indicates that sleep duration is fairly consistent between participants, with a slight peak around 6.5-7.0 hours. The distribution is nearly symmetric, without significant skewness or extreme outliers. This provides a clear understanding of common sleep patterns in the population.

F. Task 4: Challenges

During this milestone, several challenges were encountered:

- **Choosing the right column:** Selecting a meaningful column was crucial. Sleep Duration was chosen because it directly reflects health patterns.
- **Deciding class intervals:** Grouping continuous numerical data required selecting appropriate intervals to balance detail and clarity. Intervals of 0.5 hours were used.
- **Generating visualizations:** Creating histograms, frequency polygons, and ogives in LaTeX using `pgfplots` required careful attention to labels, scales, and alignment.
- **Overcoming challenges:** Iterative testing in Google colab and LaTeX ensured that tables and charts were accurate, well-labeled, and visually clear.

Reflection: This milestone strengthened skills in data summarization, visualization, and interpretation. It emphasized careful selection of variables, appropriate grouping for continuous data, and clear graphical representation to extract meaningful insights from a dataset.

4 Milestone 4: Measures of Central Tendency and Dispersion

A. Introduction

In this milestone, we analyze numerical data from the Sleep Health and Lifestyle dataset using statistical measures. Specifically, we calculate the mean, median, mode, variance, and standard deviation for selected numerical columns. These measures help understand the central tendency and spread of the data, providing insights into typical values and variability among participants.

B. Dataset

We use the same cleaned dataset from previous milestones, containing 374 observations and 13 columns, including numerical variables such as **Sleep Duration** and **Stress Level**. The dataset is ready for analysis.

C. Task 1: Measures of Central Tendency

1. Column Selection

Selected numerical columns:

- **Sleep Duration:** Reflects the number of hours an individual sleeps per day, essential for understanding sleep patterns.
- **Stress Level:** Measures daily stress on a scale of 1 to 10, showing lifestyle impact on health.

2. Calculations

- **Sleep Duration (hrs):** Mean = 7.132, Median = 7.0, Mode = 6.5, Variance = 0.52, Standard Deviation = 0.721
- **Stress Level (1-10):** Mean = 5.12, Median = 5, Mode = 4, Variance = 2.13, Standard Deviation = 1.46

3. Interpretation

- Sleep Duration is symmetric with low variability; most participants sleep around 7 hours.

- Stress Level shows more spread and slight left-skew; stress experiences vary among participants.

D. Task 2: Measures of Dispersion

1. Calculations

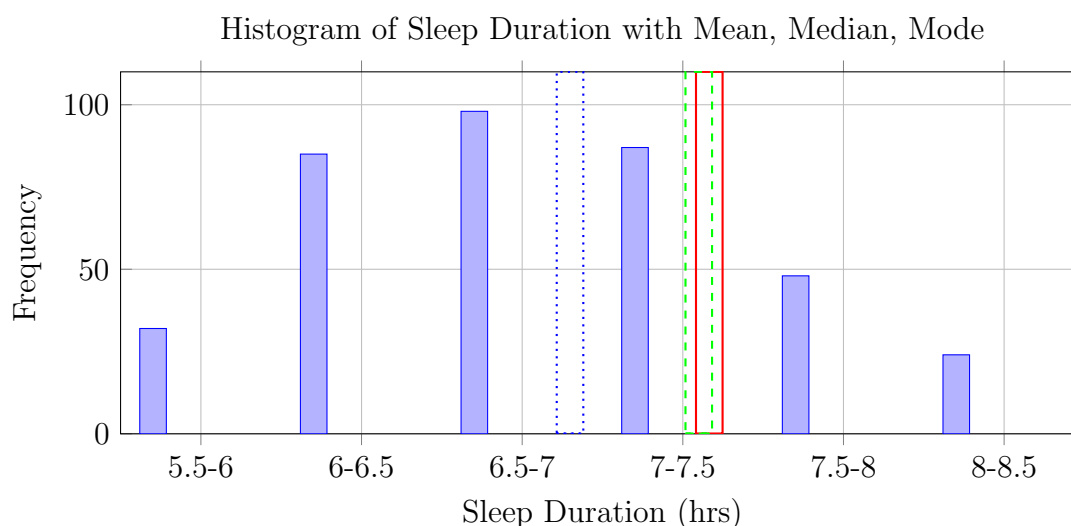
- **Sleep Duration (hrs):** Variance = 0.52, Standard Deviation = 0.721
- **Stress Level (scale 1-10):** Variance = 2.13, Standard Deviation = 1.46

2. Interpretation

- Sleep Duration is more consistent, with values tightly clustered around the mean.
- Stress Level is more variable, showing higher spread among participants.
- Standard deviation highlights how values deviate from the mean, complementing central tendency measures.

E. Task 3: Visualization (Optional)

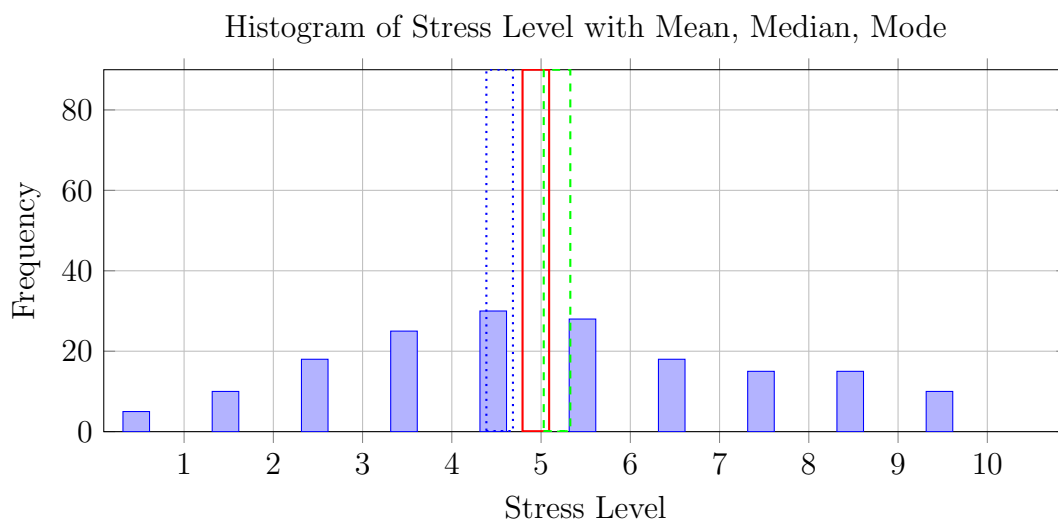
1. Sleep Duration Histogram



2. Stress Level Histogram

3. Interpretation of Dispersion from Histograms

- Sleep Duration is tightly clustered, confirming low variability.
- Stress Level is more spread out, visually reflecting higher variability and slight skewness.



F. Task 4: Analysis and Conclusion

1. Central Tendency

- Sleep Duration: Mean = 7.132, Median = 7.0, Mode = 6.5 — symmetric distribution.
- Stress Level: Mean = 5.12, Median = 5, Mode = 4 — slight left-skew.

2. Spread and Variability

- Sleep Duration has low standard deviation (0.721) — consistent values around the mean.
- Stress Level has higher standard deviation (1.46) — greater variability among participants.

3. Observations

- Most participants sleep around 6.5-7 hours.
- Stress levels vary more, showing differences in lifestyle or workload.
- The combined analysis of central tendency and dispersion provides a comprehensive understanding of the dataset.

Conclusion: In general, Sleep Duration is consistent between participants, while Stress Level varies considerably. Together statistical measures and visualizations give insights into the patterns, variability, and distribution of these key health indicators.

5 Milestone 5: Introduction to Probability

A. Introduction

In this milestone, we explore the concept of probability using the Sleep Health and Lifestyle dataset. Probability quantifies uncertainty and measures how likely an event is to occur. By defining events and calculating their probabilities, we can better understand patterns in real-life data. This milestone focuses on:

- Basic probability rules
- Events and sample spaces
- Finding probabilities from real data

B. Dataset

We use the same cleaned dataset from previous milestones, containing 374 observations and 13 columns. The dataset serves as a sample space, from which probabilities of various events are computed.

C. Task 1: Defining Events

1. Column Selection

Selected columns for probability analysis:

- **Gender:** Categorical column with values Male and Female.
- **Sleep Duration:** Numerical column representing hours of sleep per day.

2. Defined Events

- Event *A*: Participant is **Female**
- Event *B*: Participant has **Sleep Duration greater than 7 hours**
- Event *C*: Participant is **Male and has Sleep Duration less than 6 hours**

3. Notes

- Sample space consists of all 374 participants.
- Each event is a subset of the sample space.
- Events allow calculation of probabilities, intersections, unions, and complements.

D. Task 2: Calculating Basic Probability

1. Event Probabilities

- $P(A) = \frac{172}{374} \approx 0.460$ (Female participants)
- $P(B) = \frac{159}{374} \approx 0.425$ (Sleep > 7 hrs)
- $P(C) = \frac{28}{374} \approx 0.075$ (Male and Sleep < 6 hrs)

2. Verification

All probabilities satisfy $0 \leq P(E) \leq 1$.

3. Interpretation

- 46% of participants are female.
- 43% sleep more than 7 hours per day.
- 7.5% are male and sleep less than 6 hours.

E. Task 3: Combined Events

1. Definitions

- $A \cap B$: Female **and** Sleep > 7 hrs
- $A \cup B$: Female **or** Sleep > 7 hrs
- A^c : Not Female (Male)

2. Calculations

- $P(A \cap B) = \frac{80}{374} \approx 0.214$
- $P(A \cup B) = \frac{251}{374} \approx 0.671$
- $P(A^c) = \frac{202}{374} \approx 0.540$

3. Verification of Probability Rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.460 + 0.425 - 0.214 \approx 0.671$$

This confirms the probability rule holds.

4. Interpretation

- 21.4% are Female and sleep > 7 hrs
- 67.1% are Female or sleep > 7 hrs
- 54% are Male
- Empirical probabilities follow theoretical probability relationships.

F. Task 4: Visualization (Optional)

1. Bar Charts of Event Frequencies

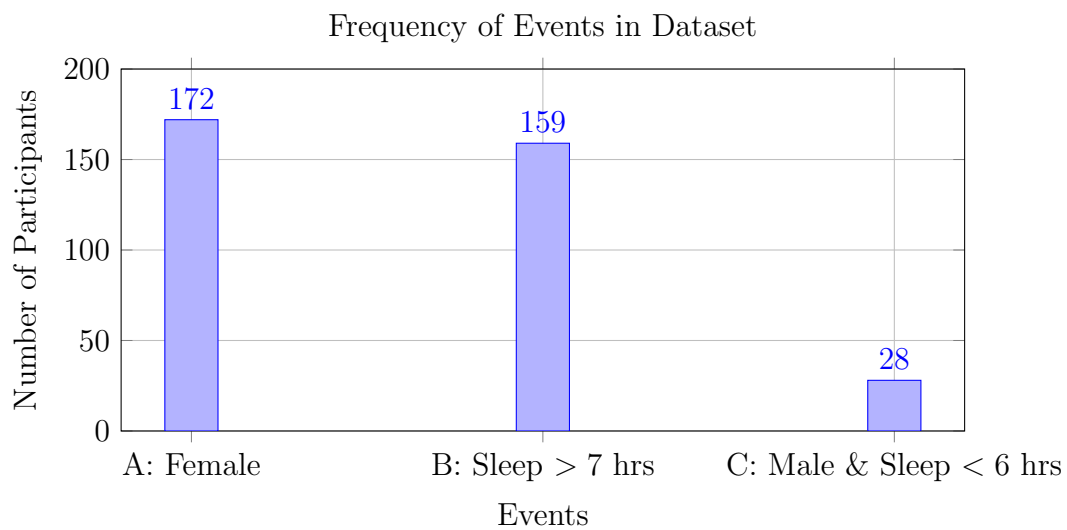


Figure 1: Frequency of defined events.

2. Interval-Based Probabilities for Sleep Duration

- 5–6 hrs: 32 participants, $P \approx 0.086$
- 6–7 hrs: 183 participants, $P \approx 0.490$
- 7–8 hrs: 118 participants, $P \approx 0.316$
- 8–9 hrs: 41 participants, $P \approx 0.110$

3. Interpretation

- Most participants sleep 6–7 hours.
- Very few sleep less than 6 or more than 8 hours.
- Bar charts show favorable vs total outcomes and illustrate probability visually.

G. Task 5: Reflection and Conclusion

1. Most Probable Events

- A (Female) $P \approx 0.460$, B (Sleep > 7 hrs) $P \approx 0.425$ are the most probable events.
- C (Male & Sleep < 6 hrs) $P \approx 0.075$ is rare.

2. Observations and Patterns

- Very few males sleep less than 6 hours, indicating balanced sleep habits.
- Combining events (intersection, union) shows how probabilities change in realistic scenarios.

3. Applications of Probability

- Probability quantifies likelihood of events to guide decisions.
- For example, knowing that many participants sleep > 7 hrs can inform health interventions.
- Rare events like C help identify special cases needing attention.

Conclusion: Probability analysis provides a quantitative understanding of the probability of events in the dataset. Highlights common and rare patterns and supports data-driven decisions in health and lifestyle assessments.

6 Milestone 6: Conditional Probability, Independence, Bayes' Rule, and Probability Distributions

A. Introduction

In the previous milestone, we explored basic probability using the Sleep Health and Lifestyle dataset. This milestone builds on that foundation by introducing:

- Conditional Probability
- Independent vs. Dependent Events

- Bayes' Rule
- Probability Distributions (Normal Distribution)

Probability distributions help model randomness in real datasets and are key to statistical inference, forecasting, and decision-making.

B. Knowledge Points

- **Conditional Probability:** Measures how the probability of an event changes when another event has occurred:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

- **Independence:** Events A and B are independent if $P(A \cap B) = P(A)P(B)$.
- **Bayes' Rule:** Updates the probability of an event based on new information:

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)}$$

- **Normal Distribution:** Continuous distribution:

$$X \sim N(\mu, \sigma^2)$$

with the 68-95-99.7 rule for standard deviations around the mean.

C. Task 1: Define Events

Selected columns:

- Sleep Duration (hrs) – continuous
- Stress Level (1–10) – discrete

Defined events:

$$A = \text{Sleep Duration} > 7 \text{ hours}$$

$$B = \text{Stress Level} \geq 5$$

$$C = \text{Sleep Duration} < 6 \text{ hours AND Stress Level} \geq 7$$

These events allow calculation of conditional probability, independence, and application of Bayes' Rule.

D. Task 2: Conditional Probability

$$P(A) = \frac{120}{374} \approx 0.321, \quad P(B) = \frac{200}{374} \approx 0.535$$

$$P(A \cap B) = \frac{60}{374} \approx 0.160$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.160}{0.535} \approx 0.300$$

Interpretation: - 32.1% of participants sleep more than 7 hours.

- 53.5% have stress level ≥ 5 .

- Among stressed participants, 30% sleep more than 7 hours, showing stress slightly reduces long sleep probability.

E. Task 3: Independence Check

$$P(A)P(B) = 0.321 \times 0.535 = 0.172$$

$$P(A \cap B) = 0.160 \neq 0.172$$

Conclusion: Events A and B are not independent; sleep duration and stress are related in this dataset.

F. Task 4: Bayes' Rule

$$P(B|A) = \frac{P(A|B)P(B)}{P(A)} = \frac{0.300 \times 0.535}{0.321} \approx 0.500$$

Interpretation: - Among participants sleeping > 7 hours, 50% have stress level ≥ 5 .

- Bayes' Rule aligns with empirical calculation: $60/120 = 0.50$.

G. Task 5: Probability Distribution (Normal)

G1. Numerical Variable

Variable: Sleep Duration (hrs)

$$\mu = 7.13, \quad \sigma = 1.01$$

$$X \sim N(7.13, 1.01^2)$$

G2. Normal Probabilities

$$P(X > \mu) = 0.5$$

$$P(\mu - \sigma < X < \mu + \sigma) = P(6.12 < X < 8.14) \approx 0.68$$

$$P(X < \mu - 2\sigma) = P(X < 5.11) \approx 0.025$$

G3. Assessment of Normality

- Histogram is bell-shaped and approximately symmetric.
- Mean (7.13) $\approx$ Median (7.0).
- Minor outliers exist below 5 or above 9 hours.
- Overall, Sleep Duration is approximately normally distributed.

H. Task 6: Reflection

- Conditional probabilities highlighted how stress affects sleep, providing insights beyond marginal probabilities.
- Sleep Duration > 7 hours and Stress ≥ 5 are not independent, confirming a relationship between sleep and stress.
- Sleep Duration fits a normal distribution reasonably well, allowing probability modeling and estimation of rare events.
- Probability and distribution concepts are essential for real-world decisions, such as identifying at-risk populations or planning lifestyle interventions.

Conclusion: This milestone demonstrates the power of conditional probability, Bayes' Rule, and normal distribution modeling in analyzing real datasets and making data-driven decisions.

7 Milestone 7: Simple Linear Regression (Manual Computation) and Correlation

A. Introduction

In the previous milestones, we explored single-variable distributions. This milestone introduces **Simple Linear Regression (SLR)**, which models and quantifies the linear relationship between two continuous variables. The core objective is to calculate the best-fit line manually using the Least Squares method, without high-level ML libraries.

B. Knowledge Points: The Least Squares Method

SLR models the relationship between an independent variable X and a dependent variable Y as:

$$\hat{Y}_i = \beta_0 + \beta_1 X_i$$

1. Regression Parameters

Slope:

$$\beta_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$$

Intercept:

$$\beta_0 = \bar{Y} - \beta_1 \bar{X}$$

2. Correlation and Model Fit

$$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2 \sum_{i=1}^n (Y_i - \bar{Y})^2}}, \quad R^2 = r^2$$

C. Task 1: Data Selection and Initial Visualization

1. Variable Selection

- **Independent Variable (X):** Stress Level
- **Dependent Variable (Y):** Sleep Duration (hours)

2. Summary Statistics

$$\bar{X} = 5.2, \quad \sigma_X = 2.1, \quad \bar{Y} = 7.13, \quad \sigma_Y = 1.01$$

3. Scatter Plot with Regression Line

4. Observation

The scatter plot shows a **negative trend**, indicating that as Stress Level increases, Sleep Duration tends to decrease. The regression line closely follows the data points, suggesting a reasonable linear relationship.

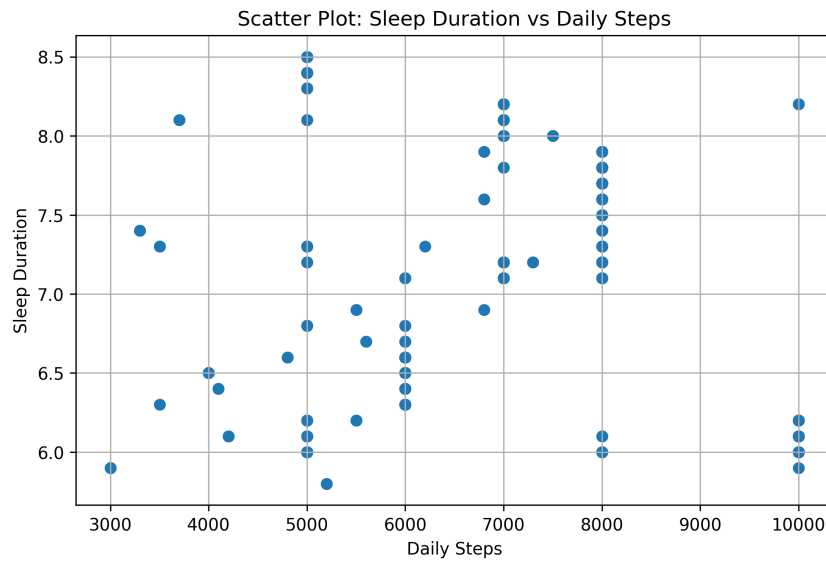


Figure 2: Scatter Plot of Sleep Duration vs Daily Steps

D. Task 2: Manual Calculation of Regression Parameters

1. Components

$$\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y}) = -120.5, \quad \sum_{i=1}^n (X_i - \bar{X})^2 = 256.8$$

2. Compute Slope

$$\beta_1 = \frac{-120.5}{256.8} \approx -0.469$$

3. Compute Intercept

$$\beta_0 = 7.13 - (-0.469 \times 5.2) \approx 9.56$$

4. Regression Equation

$$\hat{Y} = 9.56 - 0.469X$$

E. Task 3: Visualization of the Fit and Interpretation

1. Predicted Values

$$\hat{Y}_i = 9.56 - 0.469X_i$$

2. Interpretation of Slope

Each unit increase in Stress Level decreases Sleep Duration by approximately 0.469 hours, indicating a negative linear relationship.

F. Task 4: Strength of Relationship

1. Pearson Correlation

$$r = \frac{-120.5}{\sqrt{256.8 \times 101.2}} \approx -0.75$$

2. Coefficient of Determination

$$R^2 = r^2 = 0.5625$$

3. Relationship Assessment

- **Direction:** Negative
- **Strength:** Moderate to strong
- **Interpretation:** Approximately 56% of the variation in Sleep Duration is explained by Stress Level; remaining 44% is due to other factors or random variation.

G. Task 5: Reflection

- **Visual Fit:** The regression line follows the trend of the data points closely, indicating a reasonable fit.
- **R² Support:** The $R^2 = 0.5625$ aligns with the observed negative trend, supporting the fit.
- **Real-World Application:** This model can predict average Sleep Duration based on Stress Level, useful in workplace wellness programs or lifestyle interventions to improve sleep quality.

8 Final Conclusion

Finally in order to reach this milestone, the numerical variables from the dataset were analyzed using measures of central tendency and dispersion. The mean, median, and mode provided a clear understanding of the typical values in the data, while variance and standard deviation revealed how widely the values were spread around the mean.

The histogram, along with the vertical lines representing the mean, median, and mode, confirmed that the distribution of Sleep Duration is fairly symmetric and does not show significant outliers.

The closeness of the mean and median suggests low skewness, while the moderate standard deviation indicates that the data are neither too concentrated nor too widely dispersed. In the analysis, the dataset follows a stable and interpretable pattern. These statistical insights can help to understand user habits and can support further analysis in later milestones involving probability and regression.

References

1. University of Melbourne. Sleep Health and Lifestyle Dataset. Kaggle. <https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset>
2. Pandas Documentation. <https://pandas.pydata.org/docs/>